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Dyslipidemia in children and adolescents: Management

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INTRODUCTION

Dyslipidemias are disorders of lipoprotein metabolism that may result in the following abnormalities ([table 1](#)):

- High total cholesterol (TC)
- High low-density lipoprotein cholesterol
- High non-high-density lipoprotein cholesterol (non-HDL-C)
- High triglycerides (TG)
- Low HDL-C

The management of dyslipidemia in children and adolescents will be reviewed here. The definition of pediatric dyslipidemia, screening to identify children with lipid disorders in children, familial hypercholesterolemia (FH), other inherited disorders of cholesterol, pediatric prevention of adult atherosclerotic cardiovascular disease (ASCVD), and management of hypercholesterolemia in adults are reviewed separately:

- (See "[Dyslipidemia in children and adolescents: Definition, screening, and diagnosis](#)".)
- (See "[Familial hypercholesterolemia in children](#)".)

- (See ["Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia"](#).)
 - (See ["Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children"](#).)
 - (See ["Low-density lipoprotein cholesterol-lowering therapy in the primary prevention of cardiovascular disease"](#).)
 - (See ["Management of low density lipoprotein cholesterol \(LDL-C\) in the secondary prevention of cardiovascular disease"](#).)
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RATIONALE FOR INTERVENTION

The rationale for initiating lipid-lowering therapy during childhood and adolescence is based upon evidence that pediatric dyslipidemia (particularly elevated low-density lipoprotein cholesterol [LDL-C]) contributes to atherosclerosis and the development of premature atherosclerotic cardiovascular disease (ASCVD) [1]. (See ["Overview of risk factors for development of atherosclerosis and early cardiovascular disease in childhood"](#) and ["Dyslipidemia in children and adolescents: Definition, screening, and diagnosis"](#), section on 'Rationale for lipid screening'.)

Evidence establishing the long-term effectiveness of lipid-lowering interventions in children comes largely from studies in children with familial hypercholesterolemia (FH), a group at high risk for morbidity and early mortality. Lipid-lowering therapy in children with FH decelerates the atherosclerosis process, as assessed by subclinical vascular findings (eg, carotid intima-media thickness). Long-term outcome data on cardiovascular morbidity and mortality are not available for the general pediatric population; however, it is reasonable to assume, based on evidence from adult studies and limited pediatric data, that timely intervention to address dyslipidemia in childhood may decelerate the atherosclerotic process, which would prevent or delay the onset of ASCVD. (See ["Familial hypercholesterolemia in children"](#).)

The practice of initiating interventions for pediatric dyslipidemia during childhood is supported by several organizations, including the American Heart Association, the American Academy of Pediatrics, the National Lipid Association, the National Cholesterol Education

Program, and an expert panel sponsored by the National Heart, Lung, and Blood Institute [1-5]. (See '[Society guideline links](#)' below.)

Because evidence is lacking to directly link drug treatment with a reduction in ASCVD, some question the risk-benefit of pharmacologic therapy in children with dyslipidemia [6]. (See '[Adverse effects](#)' below and "[Dyslipidemia in children and adolescents: Definition, screening, and diagnosis](#)", section on '[Harms of screening](#)'.)

REFERRAL

Referral to a pediatric lipid specialist may be warranted in some cases of pediatric dyslipidemia, as described below. A pediatric lipid specialist is typically pediatric cardiologist (or an endocrinologist, gastroenterologist, or general pediatrician), who has completed additional training in lipid disorders through a senior fellowship or via professional societies. If a pediatric lipid specialist is not available locally, referral to an adult lipid specialist may be helpful, particularly for adolescents.

Referral to a pediatric lipid specialist is generally warranted for:

- Patients with known primary genetic lipid disorders (eg, familial hypercholesterolemia [FH] or other inherited disorder), particularly those not responsive to standard initial therapies ([table 2](#)). (See "[Familial hypercholesterolemia in children](#)" and "[Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia](#)".)
- **Patients with low-density lipoprotein cholesterol (LDL-C) levels ≥ 190** (4.9 mmol/L), if confirmed on repeat testing and not responsive lifestyle interventions:
 - **LDL-C levels in the range of 190 to 399 mg/dL** (4.9 to 10.3 mmol/L) that are unresponsive to dietary and lifestyle changes are likely to be due to heterozygous FH or a similar genetic lipid disorder that substantially increases the risk of premature ASCVD during young adulthood. Genetic testing may be helpful and specialist care will often be beneficial. (See "[Familial hypercholesterolemia in children](#)", section on '[Heterozygous FH \(HeFH\)](#)'.)

- **LDL-C levels ≥ 400 mg/dL (10.3 mmol/L) are likely due to homozygous FH (HoFH).** Patients with HoFH are at risk for atherosclerotic cardiovascular disease (ASCVD) in early childhood. They require genetic testing, as well as input from a pediatric cardiologist and a lipid specialist experienced in caring for children with this diagnosis. (See "[Familial hypercholesterolemia in children](#)", section on '[Homozygous FH \(HoFH\)](#)'.)
- Patients with conditions that put them at high risk for early ASCVD ([table 3](#)) who have LDL-C levels that warrant pharmacotherapy (ie, LDL-C ≥ 130 mg/dL [3.4 mmol/L]) despite optimal management of the underlying condition. (See '[Risk stratification](#)' below.)
- Patients requiring second-line pharmacologic therapy for treatment of hypercholesterolemia. (See '[Second-line agents and other therapies](#)' below.)
- **Patients with extremely elevated triglyceride (TG) levels (eg, >1000 mg/dL [11.3 mmol/L]), which raises the likelihood of a primary genetic hypertriglyceridemia.** (See '[Hypertriglyceridemia](#)' below.)

HYPERCHOLESTEROLEMIA

Risk stratification — In children, traditional cardiovascular risk factors and other specific conditions are associated with increased risk of early atherosclerotic cardiovascular disease (ASCVD). We agree with the risk stratification schema established by the American Heart Association, which categorizes these conditions and risk factors as "high-risk," "moderate-risk," or "at-risk," as summarized in the table and algorithm ([table 3](#) and [algorithm 1](#)) [1,7]. Children initially categorized in the at-risk or moderate-risk tier, based on their primary diagnosis, should be moved to a higher tier if they have additional risk factors ([algorithm 1](#)). Children with isolated hypercholesterolemia without other ASCVD risk factors or underlying conditions are generally considered "at-risk." Since high-risk conditions are uncommon in childhood, most pediatric patients with dyslipidemia will be categorized as moderate- or at-risk.

Risk factors for premature ASCVD in childhood are discussed in greater detail separately. (See

"Overview of risk factors for development of atherosclerosis and early cardiovascular disease in childhood".)

Risk-based management approach

Marked hypercholesterolemia (LDL-C >400 mg/dL [10.3 mmol/L]) — Children with low-density lipoprotein cholesterol (LDL-C) ≥ 400 mg/dL (10.3 mmol/L) should be referred to a pediatric lipid specialist and a pediatric cardiologist as soon as they are identified. LDL-C values in this range indicate that the child likely has a primary lipid disorder (eg, HoFH or severe HeFH). Children with HoFH are unlikely to respond to dietary and lifestyle modifications alone and usually require pharmacologic intervention with multiple agents. In addition, children with FH are likely to have affected family members who may benefit from testing and treatment. FH is discussed in detail separately. (See ["Familial hypercholesterolemia in children"](#).)

Young children (<10 years old) — For young children (<10 years old) with hypercholesterolemia (defined as LDL-C ≥ 130 mg/dL [3.4 mmol/L] or total cholesterol [TC] ≥ 200 mg/dL), management consists mainly of lifestyle changes. (See ["Heart-healthy lifestyle"](#) below.)

Young children with high-risk ASCVD conditions or severe primary hyperlipidemia may benefit from early pharmacotherapy. Patients with any of these conditions should be referred to a pediatric lipid specialist:

- High-risk ASCVD condition ([table 3](#))
- LDL-C levels >400 mg/dL (10.3 mmol/L), which suggest homozygous familial hypercholesterolemia (HoFH) or severe heterozygous familial hypercholesterolemia (HeFH) (see ["Familial hypercholesterolemia in children"](#))
- Strong family history of premature ASCVD

Children and adolescents (10 years and older) — Treatment of hypercholesterolemia in older children (>10 years old) and adolescents includes:

- Nonpharmacologic measures (ie, heart-healthy lifestyle changes, including dietary

modification, physical activity, weight loss, and avoidance of nicotine), which are appropriate for **all** patients with hypercholesterolemia (see '[Heart-healthy lifestyle](#)' below)

- Pharmacologic therapy (chiefly with statins), which is reserved for high-risk patients and those who do not achieve adequate response to lifestyle changes (see '[Statin therapy](#)' below)

The decision to initiate lipid-lowering medication depends upon the severity of dyslipidemia and presence of other ASCVD risk factors ([table 3](#) and [algorithm 2](#)).

The LDL-C value used in determining the need for pharmacotherapy should be based on multiple measurements rather than a single measurement. We typically base our management decisions on at least two fasting lipid profiles obtained two weeks to three months apart.

- **High-risk patients** – For patients in the high-risk category who have LDL-C ≥ 130 mg/dL (3.4 mmol/L), we suggest management with both lifestyle changes and statin therapy, started simultaneously. The treatment target once on statin therapy is LDL-C < 100 mg/dL (2.6 mmol/L). Additional details of lifestyle intervention and statin therapy are provided below. (See '[Heart-healthy lifestyle](#)' below and '[Statin therapy](#)' below.)
- **Moderate- and at-risk patients** – For patients in the moderate-risk category, initial management consists of lifestyle changes (see '[Heart-healthy lifestyle](#)' below). If the LDL-C remains ≥ 160 mg/dL (4.1 mmol/L) despite three months of lifestyle changes, we suggest initiating statin therapy. The treatment target once on statin therapy is LDL-C < 130 mg/dL (3.4 mmol/L). (See '[Statin therapy](#)' below.)
- **At-risk patients** – For patients in the at-risk category, initial management consists of lifestyle changes (see '[Heart-healthy lifestyle](#)' below). If the LDL-C remains ≥ 160 mg/dL (4.1 mmol/L) despite six months of lifestyle changes, we suggest initiating statin therapy. The treatment target once on statin therapy is LDL-C < 130 mg/dL (3.4 mmol/L). (See '[Statin therapy](#)' below.)
- **Patients without other ASCVD risk factors** – For patients with isolated

hypercholesterolemia without other ASCVD risk factors or underlying conditions, initial management consists of lifestyle changes. (See ['Heart-healthy lifestyle'](#) below.)

- **Persistent LDL-C ≥ 190 mg/dL (4.9 mmol/L) despite initial management** – Patients who continue to have severely elevated LDL-C levels (ie, ≥ 190 mg/dL [4.9 mmol/L]) despite initial lifestyle interventions should be started on statin therapy (if not already initiated) and referred to a pediatric lipid specialist. These patients are likely to have HeFH or a similar genetic lipid disorder that substantially increases the risk of premature ASCVD. (See ['Referral'](#) above and ['Statin therapy'](#) below and ["Familial hypercholesterolemia in children"](#).)

Treatment goals — LDL-C goals for the individual patient depend on the presence of associated risk factors and are generally as follows:

- For patients in the high-risk category, the treatment goal is LDL-C < 100 mg/dL (2.6 mmol/L)
- For patients in the moderate-risk and at-risk categories (including those with no additional risk factors or comorbidities), the treatment goal is LDL-C < 130 mg/dL (3.4 mmol/L)

Heart-healthy lifestyle — For all children with hypercholesterolemia, initial management includes heart-healthy lifestyle changes consisting of dietary modification, physical activity, weight loss for obese children, maximizing high-quality sleep, and avoidance of nicotine exposure.

For children with comorbidities and risk factors that place them at high risk for early ASCVD ([table 3](#)), we typically start statin therapy simultaneously with lifestyle changes (see ['Statin therapy'](#) below). For children in the moderate-risk and at-risk categories (including those with no additional risk factors), lifestyle changes are initiated first and pharmacotherapy is added if the targeted LDL-C goal is not met by three to six months.

An observational study that tracked blood lipid levels of children starting at age 9, 12, and 15 years through young adulthood demonstrated that adult lipid levels were influenced by lifestyle changes [8]. As an example, weight control and exercise were associated with improved lipid levels in individuals with pediatric dyslipidemia. In contrast, an increase in

adiposity and continuation or commencement of smoking from childhood to adulthood were associated with adult dyslipidemia.

Dietary modification — Dietary interventions can modestly improve abnormal lipid levels in healthy children and in those with dyslipidemia [1]. Dietary counseling should be tailored to the patient and his or her family. Behavior modification and motivational interviewing techniques may be helpful. Support and lifestyle modification for the whole family is usually necessary for successful change in the child. (See "[Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children](#)", section on 'Heart-healthy diet'.)

The primary dietary modifications for children with elevated LDL-C include reduced intake of saturated fat and cholesterol and increased intake of dietary fiber through fruits, vegetables, and whole grains. Dietary supplements (eg, plant sterols and stanol esters) also appear to modestly reduce LDL-C.

Heart-healthy diet — Dietary modification in children with hypercholesterolemia is carried out in a staged approach [1,3-5,7,9]:

- All children should be encouraged to eat a diet that is high in fiber from fruits, vegetables, and whole grains; high in polyunsaturated and monounsaturated fats; low in saturated fat; and devoid of trans fats ([table 4](#)). Fat should comprise approximately 30 percent of total energy intake, and saturated fats should be limited to <10 percent of total energy intake. This guidance applies to all children and is particularly important for children with dyslipidemia.
- For patients with persistent hypercholesterolemia despite appropriate dietary modification, further restricting fat intake may be reasonable (eg, limiting total fat to 25 to 30 percent of total calories and saturated fat to ≤ 7 percent of total calories). However, adherence to more restrictive diets is difficult and may foster noncompliance and stress between the child and parents/caregivers.
- We emphasize the intake of "healthier" fats (mono- and polyunsaturated fats) as a substitute for saturated and trans fats because of their beneficial effects on cardiovascular health ([table 5](#)). Dietary supplementation with plant stanols and

sterols can also be considered in this setting, as discussed below. (See '[Dietary supplements](#)' below.)

Consultation with a registered dietitian is advised for implementation of dietary changes.

Based on the available data, a diet low in saturated fats and high in dietary fiber appears to modestly lower LDL-C levels in children with hypercholesterolemia [10,11]. However, for children with FH or LDL-C >190 mg/dL (4.9 mmol/L), diet modification alone is rarely sufficient to reach target levels for LDL-C. (See "[Familial hypercholesterolemia in children](#)", section on '[Management](#)'.)

In a randomized trial in 663 prepubertal children with LDL-C levels between the 80th and 98th percentile for age and sex, those assigned to a restricted diet (total fat intake to 28 percent of total calories, saturated fat <8 percent of total calories, and TC to 75 mg/1000 kcal per day) had a larger reduction of LDL-C levels over a three-year period compared with the control group (15.4 versus 11.9 mg/dL [0.40 versus 0.31 mmol/L]) [11]. There were no statistically significant differences in TC, high-density lipoprotein cholesterol (HDL-C), and TG levels between the two groups. In another study of prepubertal children with FH, following a diet low in total fat (23±5 percent of total calories), saturated fat (8±2 percent of total calories), and cholesterol (67±28 mg/1000 kcal) for one year resulted in reductions in TC and LDL-C levels by 4 and 5.5 percent, respectively [10]. There was no difference in HDL-C or TG levels.

Larger studies in healthy children have also demonstrated that a diet low in saturated fats is associated with improvements in lipid profiles and other cardiovascular measures [12-15]. These data are presented separately. (See "[Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children](#)", section on '[Heart-healthy diet](#)'.)

Dietary supplements — Dietary supplementation with plant stanols and sterols appears to enhance a reduction in LDL-C in conjunction with a low-saturated fat diet [1]. The benefit of other supplements (eg, fiber) to reduce LDL-C is less clear. The following is a summary of these supplements, which are discussed in greater detail separately. (See "[Lipid management with diet or dietary supplements](#)", section on '[Our approach](#)'.)

- **Plant stanols and sterols** – These compounds are found naturally in fruits, vegetables,

vegetable oils, nuts, and seeds and are additives in a number of other foods, including certain margarines, orange juice, yogurt drinks, cereal bars, and dietary supplements. In adults, supplementation with these compounds has been shown to reduce dietary cholesterol by 5 to 10 percent. One study in healthy children showed that a mean intake of 1.8 g/day of plant sterol contained in a margarine product reduced serum LDL-C by 8 percent [16]. In children with FH, clinical trials have shown that both stanols and sterol esters decrease TC and LDL-C levels [17-19]. Although stanols and sterols reduce serum LDL-C levels, there are no data demonstrating a reduction in the risk of early ASCVD [20,21]. Stanols and sterols decrease absorption of fat-soluble vitamins and [beta-carotene](#). Children taking these supplements should therefore take a daily multivitamin.

- **Fiber** – It is unclear whether increased fiber intake from supplements reduces serum LDL-C [1,3]. Fiber is thought to bind with cholesterol within bile acids, thus removing it from the enterohepatic circulation. In our practice, we encourage children to consume fiber from dietary sources (eg, fruit, vegetables, whole grains) rather than fiber supplements. Appropriate daily fiber intake in this setting is 6 g for children ages 2 to 12 years and 12 g for children ages 12 years and older.
- **Ineffective supplements** – Other supplements, such as garlic, are not effective. Red yeast rice extract, which contains monacolins with statin-like activity, has been shown to lower cholesterol in adults. However, we do not recommend its use, because the amount of monacolins a child would be exposed to is unknown and unregulated. (See "[Lipid management with diet or dietary supplements](#)", section on 'Supplements that we do not recommend' and "[Statin muscle-related adverse events](#)", section on 'Red yeast rice'.)

Omega-3 fatty acids, as found in fish oil supplements, are not recommended for children with hypercholesterolemia, as they may increase LDL-C levels. However, they are sometimes used to treat patients with hypertriglyceridemia, as discussed below. (See '[Hypertriglyceridemia](#)' below and "[Lipid management with diet or dietary supplements](#)", section on 'Dietary supplements that may be of benefit'.)

Physical activity — Evidence in children and adults demonstrates that daily vigorous activity and reduction in sedentary behavior are associated with improved lipid profiles and

lower risk of ASCVD. It is unclear whether this is an independent association or whether it is mediated by weight loss. Children with dyslipidemia should participate in at least as much daily physical activity as recommended for the general pediatric population ([table 6](#)). Physical activity and cardiovascular health in children and adults are discussed separately. (See "[Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children](#)", section on 'Physical activity' and "[Exercise and fitness in the prevention of atherosclerotic cardiovascular disease](#)".)

Weight loss — For children with obesity and dyslipidemia, weight loss can result in substantial improvement in lipid values [22-24]. Management of childhood obesity is discussed separately. (See "[Prevention and management of childhood obesity in the primary care setting](#)".)

Avoiding nicotine exposure — Counseling to prevent smoking initiation is an important aspect of routine pediatric primary care for all children and adolescents and particularly important for those with dyslipidemia. For children and adolescents who either are current smokers or have significant secondhand exposure, counseling about smoking cessation should be included in the discussion of heart-healthy lifestyle changes. These issues are discussed in greater detail separately. (See "[Prevention of smoking and vaping initiation in children and adolescents](#)" and "[Management of smoking and vaping cessation in adolescents](#)" and "[Secondhand smoke exposure: Effects in children](#)".)

Statin therapy — The decision to initiate lipid-lowering medication (chiefly with statins) depends upon the age of the child, severity of dyslipidemia, and presence of other ASCVD risk factors ([algorithm 2](#)), as discussed above. (See '[Risk-based management approach](#)' above.)

Dosing and efficacy — Statins are the preferred first-line agents for treatment of hypercholesterolemia in children who meet criteria for pharmacotherapy. Several different formulations of statin therapy are available. [Lovastatin](#), [simvastatin](#), [pravastatin](#), [rosuvastatin](#), and [atorvastatin](#) are approved by the US Food and Drug Administration for use in children ([table 7](#)). Pravastatin, atorvastatin, simvastatin, and lovastatin are available in generic forms. The initial choice of agent is generally based on potential drug interactions, price, and patient preference. Treatment is initiated at the lowest dose ([table 7](#)), which is

given once a day, usually at bedtime because most LDL-C synthesis occurs during nighttime hours. If needed, the dose is increased to meet the goals of therapy. (See '[Treatment goals](#)' above and '[Dose titration](#)' below.)

Several clinical trials in children with FH demonstrated that statin therapy, compared with placebo, decreases serum LDL-C without significant adverse side effects over many years of follow-up [25-36]. On average, the reduction in serum LDL-C demonstrated in these trials was approximately 25 to 40 mg/dL (0.65 to 1 mmol/L) [29]. Measures of preclinical atherosclerosis (eg, carotid intima-media thickness, flow-mediated dilation) suggest statin therapy slows the progression of subclinical atherosclerosis [25-27,37] and that these benefits of early treatment are sustained over time [36,38]. In one study, individuals with FH who started statin therapy during childhood had a lower rate of cardiovascular events in adulthood compared with their affected parents [36]. The evidence supporting use of statins in children with FH is discussed in greater detail separately. (See "[Familial hypercholesterolemia in children](#)", section on '[Management of HeFH](#)'.)

The efficacy and safety of statin therapy in pediatric patients is also supported by studies of children who have undergone cardiac transplantation [39,40]. In two studies (one retrospective and one a prospective open-label clinical trial), statin therapy was associated with a lower incidence of cardiac transplant vasculopathy and lower mortality. In the prospective study, no severe adverse effects of statin therapy were observed during eight years of follow-up [40].

Statins are the most common agents used in the treatment of hypercholesterolemia in adults and are the class of drugs in which benefits to ASCVD morbidity and mortality have been most consistently demonstrated. (See "[Low-density lipoprotein cholesterol-lowering therapy in the primary prevention of cardiovascular disease](#)".)

Adverse effects — The potential benefits of ASCVD risk reduction with statin therapy must be weighed against the risk of potential adverse effects from the drug itself [6,41,42]. **Side effects with statins are rare and include myopathy, new-onset type 2 diabetes mellitus, and hepatic enzyme elevation.** In pediatric clinical trials, rates of side effects with statin therapy were low and adherence to statin therapy was generally good [25,27,29-35,43]. (See "[Statins: Actions, side effects, and administration](#)", section on '[Side effects](#)'.)

Side effects are rare in clinical practice [44]. Side effects of statins are more likely to occur at higher doses and in patients taking other medications (particularly [cyclosporine](#), azole antifungal agents, and other medications and foods [eg, grapefruit] that impact the cytochrome P450 system ([table 8](#))). It is important to review drug interactions prior to initiating therapy (specific drug interactions may be determined by using [Lexicomp drug interactions](#)).

Adolescent females should be counseled about the possibility of drug teratogenicity and appropriate contraceptive methods while receiving statin therapy. In addition, use of statins by breastfeeding mothers is discouraged. (See "[Statins: Actions, side effects, and administration](#)", section on '[Risks in pregnancy and breastfeeding](#)' and '[Second-line agents and other therapies](#)' below.)

Although data on long-term statin use in adults suggest a safe profile, the long-term safety of statin therapy initiated in childhood is uncertain [6,45]. Concerns have been raised about the theoretical risk of altering the steroid synthesis pathway by using statins in developing children; however, these concerns have not been supported in the literature and must be balanced against risks of early atherosclerotic disease. The results of a 10-year follow-up study of early statin initiation in children with FH showed no adverse effect on growth or development [28,38].

Bile acid sequestrants may be considered an alternate option for patients in whom there is a preference for avoiding a statin. (See '[Second-line agents and other therapies](#)' below.)

Monitoring — We suggest the following baseline laboratory evaluation prior to initiating statin therapy:

- Fasting lipid profile
- Serum creatine kinase
- Serum alanine aminotransferase
- Blood glucose and hemoglobin A1c levels
- Pregnancy test, if clinically indicated

Fasting lipid profile, creatine kinase, alanine aminotransferase, glucose, and hemoglobin A1c are repeated four weeks after starting therapy to determine the response to treatment and

to assess for adverse effects ([algorithm 3](#)). Once on stable therapy, lipid levels, glucose, and hemoglobin A1c are repeated every six months. Ongoing monitoring of growth and other measures of general and cardiovascular health (eg, blood pressure) should also occur at each visit. Repeated testing of liver function tests and creatine kinase in asymptomatic patients is generally not necessary. However, testing should be performed if the patient has concerning symptoms (eg, muscle aches or weakness) or other comorbidities (eg, liver disease). Providers should be aware that estrogen-containing contraceptives can increase lipid levels. We generally recheck a fasting lipid profile approximately one month after starting an estrogen-containing contraceptive to assess its effect.

Dose titration — Statin therapy is aimed at achieving an LDL-C value <130 mg/dL (3.35 mmol/L) for patients in the moderate- and at-risk categories and a value <100 mg/dL (2.59 mmol/L) for patients in the high-risk category. As the evidence supporting a particular LDL-C goal in youth is scant, decisions about the use of higher-dose statins can be made together with the patient/family, balancing the risk of adverse effects with potential benefits related to the severity of family history, baseline LDL-C levels, and the presence of additional ASCVD risk factors. (See '[Risk stratification](#)' above.)

Titration of statin therapy is accomplished as follows ([algorithm 3](#)):

- If the LDL-C goal is met and there are no laboratory abnormalities, therapy is continued at the same dose.
- If the LDL-C goal is not met, the dose is increased (usually by an increment of 10 to 20 mg), and a repeat LDL-C level is obtained in four weeks, including repeat laboratory evaluation.
- If goals are not met after increasing the dose, but the treatment is well tolerated, one of the following options are followed:
 - The dose of statin may be increased further until the target is met, the maximum dose is reached ([table 7](#)), or there is evidence of toxicity. Laboratory testing is reassessed after each dose change.
 - Another drug (eg, [ezetimibe](#) or a bile acid sequestrant) may be added to statin

therapy under the direction of a pediatric lipid specialist. Care must be taken if fibrate is added to a statin because this may increase the risk of side effects, particularly muscle toxicity, depending on the additional agent added. (See ['Adverse effects'](#) above and ['Second-line agents and other therapies'](#) below.)

- If laboratory abnormalities develop while on statin therapy, the drug may be stopped and the laboratory testing repeated in two weeks, or sooner depending on the severity. When the abnormalities resolve, the drug may be restarted with close monitoring. For patients with adverse symptoms possibly related to the drug, it is most helpful to obtain laboratory testing while the symptoms are ongoing. If laboratory abnormalities do not appear to be statin-related, it may be reasonable to simply repeat the laboratory tests without stopping the statin.

Patients who fail to achieve goal values of LDL-C despite lifestyle changes and pharmacologic therapy should be referred to a specialist with expertise in managing pediatric dyslipidemia, as other interventions may be warranted. (See ["Treatment of drug-resistant hypercholesterolemia"](#).)

Second-line agents and other therapies — If treatment with the maximal-tolerated dose of statin is not successful in achieving the LDL-C goal or if it is not tolerated, a number of other agents are available with varying levels of supporting evidence. Second-line lipid-altering agents encompass several classes of drugs that differ with respect to mechanism of action and to the degree and type of lipid lowering. These include cholesterol absorption inhibitors (eg, [ezetimibe](#)), bile acid sequestrants, fibric acid derivatives, and [niacin](#) ([table 7](#)). Other therapies include proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors (eg, [evolocumab](#) and [alirocumab](#)), [evinacumab](#), [bempedoic acid](#), LDL apheresis, and [lomitapide](#).

When a second agent is required, our practice is to use [ezetimibe](#) because of its safety profile and demonstrated efficacy in adult patients and because it lacks bothersome side effects, which commonly occur with other second-line agents (eg, bile acid sequestrants). However, the choice of second agent should be individualized to the patient and should generally be done in consultation with a pediatric lipid specialist.

- **Cholesterol absorption inhibitors (eg, [ezetimibe](#))** – Ezetimibe is a lipid-lowering agent

that prevents intestinal absorption of cholesterol and plant sterols. Ezetimibe can be useful in children and adolescents with FH or other high-risk factors for premature ASCVD who are not able to reach LDL-C treatment goals on high-intensity statin therapy.

In combination with statins, [ezetimibe](#) further lowers serum LDL-C and, in adults, improves cardiovascular outcomes without altering the side effect profile [46,47]. (See "[Low-density lipoprotein cholesterol lowering with drugs other than statins and PCSK9 inhibitors](#)", section on 'Ezetimibe'.)

Limited data are available on the use of [ezetimibe](#) in children. In randomized controlled trial evaluating monotherapy with ezetimibe (10 mg per day) compared with placebo in 138 children with hyperlipidemia (chiefly, patients with heterozygous FH), treatment with ezetimibe resulted in greater reductions in LDL-C (28 versus 1 percent) and TC (21 versus 0.2 percent) [48]. Side effects were mild and were comparable between the two groups; there were no serious drug-related events. In a small clinical trial of adolescent patients with FH, the combination administration of ezetimibe and a statin resulted in greater reductions of LDL-C levels than did treatment with a statin alone [49].

- **Bile acid sequestrants** – Bile acid sequestrants are not as effective as statins in lowering LDL-C and have adverse side effects that result in poor compliance (eg, constipation and bloating) [50]. For these reasons, bile acid sequestrants are used relatively infrequently. However, they may be useful in combination with a statin for patients who fail to meet target LDL-C levels [51]. The sequestrants are extremely safe as they are not absorbed systemically but remain in the gut and are excreted along with the bile-containing cholesterol. For patients with severe LDL-C elevations (heterozygous FH), the sequestrants can effectively lower LDL-C as much as 10 to 20 percent. They can be used in patients who prefer to avoid statins, although they may not achieve sufficient LDL lowering to reach goal. Their use is hampered by bloating, constipation, and altered absorption of other medications, necessitating attention to timing of other medications over the course of the day. (See "[Low-density lipoprotein cholesterol lowering with drugs other than statins and PCSK9 inhibitors](#)", section on 'Bile acid sequestrants'.)

- **Fibric acid derivatives** – Fibric acid derivatives (eg, [gemfibrozil](#), [fenofibrate](#)) are used primarily for treating severe hypertriglyceridemia and are discussed in more detail below. (See '[Pharmacotherapy](#)' below.)
- **Niacin** – Niacin is particularly effective in raising HDL-C, increasing levels by 20 to 30 percent [52]. However, compliance is difficult due to frequent side effects (eg, rash, flushing, and headaches), and adult primary prevention trials have not shown a ASCVD benefit for niacin. Pediatric data are sparse. As a result, niacin is rarely used. Niacin comes in several different formulations, including various over-the-counter preparations, so care should be taken with dosing. The prescription formulation is the most convenient for use. (See "[Low-density lipoprotein cholesterol lowering with drugs other than statins and PCSK9 inhibitors](#)", section on '[Nicotinic acid \(niacin\)](#)'.)
- **PCSK9 inhibitors** – PCSK9 inhibitors (eg, [evolocumab](#) and [alirocumab](#)) are human monoclonal antibodies that bind to PCSK9, a protein that plays a key role in regulating plasma LDL-C levels. In the United States, evolocumab is approved for use in patients aged ≥ 10 years with heterozygous or homozygous FH [53]; alicumab is approved only for use in adult patients. In Europe, evolocumab is approved for use in patients ≥ 12 years old with homozygous FH [54]. (See "[PCSK9 inhibitors: Pharmacology, adverse effects, and use](#)" and "[Familial hypercholesterolemia in children](#)", section on '[Management](#)'.)

PCSK9 inhibitors appear to have a good safety and side effect profile; however, they have the disadvantage of requiring injection for administration. The use of these agents in pediatric patients is limited to the rare patient who does not respond sufficiently or cannot tolerate oral agents and individuals with phenotypic homozygous FH (ie, compound heterozygotes). (See "[Familial hypercholesterolemia in children](#)", section on '[Management](#)'.)

In a randomized trial involving 157 adolescent patients with heterozygous FH, [evolocumab](#) reduced LDL-C levels by approximately 40 percent relative to placebo [55]. Clinical trials in adult patients have demonstrated similar reductions in LDL-C and reductions in ASCVD events. These data are discussed separately. (See "[PCSK9 inhibitors: Pharmacology, adverse effects, and use](#)", section on '[Clinical effect](#)'.)

[Inclisiran](#) is a long-acting agent that is given via subcutaneous injection. It is a small interfering ribonucleic acid molecule that reduces the synthesis of PCSK9 through gene silencing. It is approved for use in adult patients with heterozygous FH. Pediatric trials are ongoing. (See "[PCSK9 inhibitors: Pharmacology, adverse effects, and use](#)", section on '[Small interfering RNA \(inclisiran\)](#)'.)

- **Bembedoic acid** – Bembedoic acid inhibits an enzyme upstream of 3-hydroxy-3methylglutaryl-CoA reductase (the target of statins). This in turn decreases cholesterol biosynthesis, leading to an upregulation in LDL receptors to increase LDL-C clearance. Bembedoic acid is approved for use in adults with FH. (See "[Low-density lipoprotein cholesterol lowering with drugs other than statins and PCSK9 inhibitors](#)", section on '[Bempedoic acid](#)'.)
- [Evinacumab](#) – Patients with homozygous FH who have virtually absent (null-null) or impaired (non-null) LDL-receptor activity are not responsive to PCSK9 inhibitors. A newer agent showing promise for these patients is evinacumab, which is a monoclonal antibody against *ANGPTL3* [56]. In clinical trials involving adult and adolescent patients with homozygous FH, evinacumab markedly reduced LDL-C [57,58]. (See "[Low-density lipoprotein cholesterol lowering with drugs other than statins and PCSK9 inhibitors](#)", section on '[Monoclonal antibody against ANGPTL3](#)'.)
- [Lomitapide](#) – Lomitapide is another agent used chiefly in patients with homozygous FH or in adults with very severe heterozygous FH. Unlike PCSK9 inhibitor therapy, it is given orally and hence does not require injections. However, lomitapide has a greater risk of hepatotoxicity, and with the advent of other medications, this agent is rarely used in children. It is discussed in greater detail separately. (See "[Treatment of drug-resistant hypercholesterolemia](#)".)
- **LDL apheresis** – For patients at very high risk for serious cardiovascular events (eg, patients with homozygous FH) who cannot reach their LDL-C goals with standard lipid-lowering therapy, LDL apheresis is an effective method to lower circulating levels of LDL-C. LDL apheresis is discussed in greater detail separately. (See "[Treatment of drug-resistant hypercholesterolemia](#)", section on '[LDL apheresis](#)'.)

HYPERTRIGLYCERIDEMIA

Hypertriglyceridemia is defined as follows (see ["Dyslipidemia in children and adolescents: Definition, screening, and diagnosis"](#), section on 'Definition of pediatric dyslipidemia'):

- Children <10 years: Triglyceride (TG) level ≥ 100 mg/dL (1.1 mmol/L)
- Children and adolescents 10 to 19 years: TG level ≥ 130 mg/dL [1.5 mmol/L])

Clinical significance — It is important to recognize that hypertriglyceridemia by itself is not a major risk factor for atherosclerotic cardiovascular disease (ASCVD). Hypertriglyceridemia is associated with other conditions that increase the risk of ASCVD, particularly type 2 diabetes and obesity. In addition, hypertriglyceridemia is often associated with elevated atherosclerotic lipid particles as represented by elevated non-high-density lipoprotein cholesterol (non-HDL-C) or apolipoprotein B levels.

Treatment of hypertriglyceridemia primarily focuses on promoting healthy eating habits and encouraging physical activity and weight loss rather than treating the triglyceride (TG) value per se. However, pharmacology may be warranted in patients with very elevated levels to reduce the risk of pancreatitis. (See ['Lifestyle modification'](#) below and ['Pharmacotherapy'](#) below.)

Patients with extremely elevated TG levels (ie, >1000 mg/dL [11.3 mmol/L]) generally warrant specialty care since these levels raise the possibility of a primary genetic hypertriglyceridemia. Pharmacotherapy may or may not be effective at these levels depending on the cause. (See ['Referral'](#) above.)

Lifestyle modification — Lifestyle interventions for the management of hypertriglyceridemia are aimed primarily at achieving weight loss and diet modification. This includes [\[1,7\]](#):

- Encouraging physical activity (see ['Physical activity'](#) above and ["Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children"](#), section on 'Physical activity' and ["Exercise and fitness in the prevention of atherosclerotic cardiovascular disease"](#))

- Dietary changes, including reduced intake of carbohydrates, particularly simple carbohydrates, reduced intake of saturated fats, and, for obese children, calorie restriction
- Other strategies to promote weight loss (see ["Prevention and management of childhood obesity in the primary care setting"](#))

Studies in adults and several pediatric reports have shown that a reduction in carbohydrates, particularly simple carbohydrate intake, in conjunction with weight loss and increased physical activity, results in reduced TG levels [59,60]. (See ["Hypertriglyceridemia in adults: Management"](#), section on 'Treatment goals'.)

Pharmacotherapy — Pharmacotherapeutic options for children with elevated TG are limited, and the main interventions remain diet modification, increased intake of omega-3 fatty acids, increased physical activity, and weight loss. (See ["Lifestyle modification"](#) above.)

Pharmacologic treatment for pediatric hypertriglyceridemia should be initiated in consultation with a pediatric lipid specialist. We generally use the following indications for initiating pharmacologic treatment [7]:

- Patients with severely elevated TG levels (ie, TG ≥ 600 to 1000 mg/dL [6.8 to 11.3 mmol/L]) since these levels are associated with risk of pancreatitis. (See ["Hypertriglyceridemia-induced acute pancreatitis"](#).)
- Patients who have primary hypertriglyceridemia with average TG levels >400 mg/dL (4.5 mmol/L).
- Patients with both hypertriglyceridemia and hypercholesterolemia (ie, TG 150 to 399 mg/dL [1.7 to 4.5 mmol/L] **and** non-high-density lipoprotein cholesterol [non-HDL-C] ≥ 145 mg/dL [≥ 3.7 mmol/L]) that persists despite appropriate initial therapy (including lifestyle modification and/or optimal statin therapy).

Drug therapy choices include statins, high-dose marine omega-3 fatty acid supplementation, fibrates, or [niacin](#). Niacin is rarely used because of unacceptable side effects and limited evidence of benefit [52].

- **Statins** – Statins are used if hypertriglyceridemia is accompanied by high levels of atherogenic particles as represented by elevated non-HDL-C or apolipoprotein B. (See ['Statin therapy'](#) above.)
- **Marine omega-3 fatty acids** – We use high-dose marine omega-3 fatty acid therapy (2 to 4 g per day) for treating patients with severely elevated TG levels that may lead to pancreatitis (ie, TG >600 to 1000 mg/dL [6.8 to 11.3 mmol/L]). We typically use [icosapent ethyl](#) in this setting. It can be given in conjunction with fibrate or statin therapy or as a substitute if fibrates are not tolerated.

Limited data are available on the effectiveness of marine omega-3 fatty acid supplementation in pediatric patients. Two small studies evaluated treatment with an omega-3 fatty acid supplement ([omega-3-acid ethyl esters](#)) in adolescents with hypertriglyceridemia and did not find any significant reduction in TG levels compared with placebo [61,62]; however, the studies were likely underpowered.

There are no pediatric studies of [icosapent ethyl](#) for triglyceride lowering. Use of this agent is based on evidence from studies in adults with hypertriglyceridemia. Studies in adults have demonstrated that high-dose marine omega-3 fatty acid supplementation lowers serum TG levels and may reduce the risk of ASCVD events in select at-risk populations. These data are discussed separately. (See ["Hypertriglyceridemia in adults: Management"](#), section on ["Marine omega-3 fatty acids"](#).)

- **Fibrates** – Fibric acid derivatives (eg, [gemfibrozil](#), [fenofibrate](#)) are used for treating severe or refractory hypertriglyceridemia in children ([table 7](#)). These agents raise HDL-C and lower TG levels. Fibrates are usually reserved for the rare child with severe TG elevations (most useful for TG levels >1000 mg/dL [11.3 mmol/L] but may be considered for TG levels 400 to 1000 mg/dL [6.8 to 11.3 mmol/L]) and are most often used in older adolescents.

Data are limited on the use of fibric acid derivatives in children [1,63]. In one report from a tertiary center that reviewed its management of 76 children with hypertriglyceridemia (53 patients with primary and 13 with adiposity-related hypertriglyceridemia), TG levels decreased with the use of fibrates, did not change with

the use of statins, and increased with the use of bile acid-binding resins [64].

In our experience, fibric acid derivatives are generally well-tolerated. The risk of myopathy and rhabdomyolysis is increased when they are used in combination with statins or in patients with renal insufficiency. The need for simultaneous use of both statin and fibrate therapy in childhood is rare, and they should be administered under the supervision of a pediatric lipid specialist.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Lipid disorders and atherosclerosis in children](#)".)

SUMMARY AND RECOMMENDATIONS

- **Rationale for treatment** – Management of dyslipidemia in children and adolescents is based on the rationale that early identification and control of pediatric dyslipidemia will reduce the risk and severity of premature atherosclerotic cardiovascular disease (ASCVD) in adulthood. (See '[Rationale for intervention](#)' above.)
- **Heart-healthy lifestyle changes** – For all children with dyslipidemia, management includes heart-healthy lifestyle changes, consisting of dietary modification ([table 4](#)), physical activity ([table 6](#)), weight loss for obese children, and avoidance of nicotine exposure. (See '[Heart-healthy lifestyle](#)' above and "[Dietary recommendations for toddlers, preschool, and school-age children](#)".)
- **Risk-based management** – Treatment for hypercholesterolemia (defined as low-density lipoprotein cholesterol [LDL-C] level ≥ 130 mg/dL [3.4 mmol/L] or total cholesterol [TC] ≥ 200 mg/dL [5.2 mmol/L]) depends upon the age of the child, severity of dyslipidemia, and presence of other ASCVD risk factors ([table 3](#) and [algorithm 2](#)) (see '[Risk-based management approach](#)' above):

- **Marked hypercholesterolemia (eg, LDL-C $\geq$ 400 mg/dL [10.3 mmol/L])** – Children with hypercholesterolemia in this range should be referred to a pediatric lipid specialist as it is likely that they have a severe primary hyperlipidemia (eg, homozygous familial hypercholesterolemia [HoFH]). (See ['Referral'](#) above and ["Familial hypercholesterolemia in children"](#).)
- **Children <10 years old** – For children <10 years old who are in the moderate- or at-risk categories ([table 3](#)), management consists mainly of lifestyle changes (see ['Heart-healthy lifestyle'](#) above). Children <10 years who are in the high-risk category should be referred to a pediatric lipid specialist. (See ['Referral'](#) above.)
- **Children >10 years old and adolescents** – Our suggested approach to managing hypercholesterolemia (LDL-C $\geq$ 130 mg/dL [3.4 mmol/L]) in pediatric patients $\geq$ 10 years old according to their risk category ([table 3](#)) is as follows ([algorithm 2](#)) (see ['Children and adolescents \(10 years and older\)'](#) above):
 - High-risk – For patients in the high-risk category who have, we suggest initial treatment with statin therapy in conjunction with lifestyle changes rather than lifestyle changes alone (**Grade 2B**). The treatment target once on statin therapy is LDL-C <100 mg/dL (2.6 mmol/L). (See ['Heart-healthy lifestyle'](#) above and ['Statin therapy'](#) above.)
 - Moderate-risk – For patients in the moderate-risk category, initial management consists of lifestyle changes (see ['Heart-healthy lifestyle'](#) above). If the LDL-C remains $\geq$ 160 mg/dL (4.1 mmol/L) despite three months of lifestyle changes, we suggest initiating statin therapy (**Grade 2C**). The treatment target once on statin therapy is LDL-C <130 mg/dL (3.4 mmol/L). (See ['Statin therapy'](#) above.)
 - At-risk – For patients in the at-risk category, initial management consists of lifestyle changes (see ['Heart-healthy lifestyle'](#) above). If the LDL-C remains $\geq$ 160 mg/dL (4.1 mmol/L) despite six months of lifestyle changes, we suggest initiating statin therapy (**Grade 2C**). The treatment target once on statin therapy is LDL-C <130 mg/dL (3.4 mmol/L). (See ['Statin therapy'](#) above.)
 - Patients without other ASCVD risk factors – For patients with isolated

hypercholesterolemia without other ASCVD risk factors or underlying conditions, initial management consists of lifestyle changes. (See '[Heart-healthy lifestyle](#)' above.)

- Persistently elevated LDL-C – Patients who continue to have LDL-C ≥ 190 mg/dL (4.9 mmol/L) despite initial lifestyle interventions should be referred to a pediatric lipid specialist. (See '[Referral](#)' above.)

- **Statin dosing, titration, and monitoring** – Several different formulations of statin therapy are available ([table 7](#)). The initial choice of agent is generally based on potential drug interactions, price, and patient preference.

Treatment is initiated at the lowest dose and is given once a day, usually at bedtime. The dose is subsequently titrated to achieve the goal LDL-C value according to the child's risk category (<100 mg/dL [2.6 mmol/L] for high-risk patients; <130 mg/dL (3.36 mmol/L) for children in the moderate- and at-risk categories) ([algorithm 3](#)). (See '[Dosing and efficacy](#)' above and '[Dose titration](#)' above and '[Treatment goals](#)' above.)

Side effects with statins are rare and include myopathy, new-onset type 2 diabetes mellitus, and hepatic enzyme elevation. Side effects are more likely at higher doses and in patients taking other medications. Laboratory monitoring for side effects includes serum creatine kinase and serum alanine aminotransferase, which are obtained at baseline and then at four weeks after starting therapy ([algorithm 3](#)). (See '[Adverse effects](#)' above and '[Monitoring](#)' above.)

- **Management of hypertriglyceridemia** – For children with hypertriglyceridemia, interventions are aimed primarily at achieving weight loss and diet modification. Hypertriglyceridemia by itself is not a major ASCVD risk factor. Drug therapy may be warranted in severe cases to prevent pancreatitis. Statins, high-dose marine omega-3 fatty acid supplementation, and/or fibrates (eg, [gemfibrozil](#), [fenofibrate](#)) are typically used in this setting, in conjunction with aggressive lifestyle modification and management of coexisting conditions and risks (eg, diabetes, thyroid disease, renal insufficiency). (See '[Hypertriglyceridemia](#)' above.)

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Topic 17048 Version 39.0

GRAPHICS

Acceptable, borderline, and abnormal plasma lipid, lipoprotein, and apolipoprotein concentrations for children and adolescents

Category	Acceptable mg/dL (mmol/L)	Borderline mg/dL (mmol/L)	High mg/dL (mmol/L)
TC	<170 (4.4)	170 to 199 (4.4 to 5.2)	≥200 (5.2)
LDL-C	<110 (2.8)	110 to 129 (2.8 to 3.3)	≥130 (3.4)
Non-HDL-C	<120 (3.1)	120 to 144 (3.1 to 3.7)	≥145 (3.8)
ApoB	<90 (2.3)	90 to 109 (2.3 to 2.8)	≥110 (2.8)
TG			
• 0 to 9 years	<75 (0.8)	75 to 99 (0.8 to 1.1)	≥100 (1.1)
• 10 to 19 years	<90 (1 mmol/L)	90 to 129 (1 to 1.5)	≥130 (1.5)
Category	Acceptable mg/dL (mmol/L)	Borderline mg/dL (mmol/L)	Low mg/dL (mmol/L)
HDL-C	>45 (1.2)	40 to 45 (1 to 1.2)	<40 (1)
ApoA-1	>120 (3.1)	115 to 120 (3 to 3.1)	<115 (3)

Values for plasma lipid and lipoprotein levels are from the National Cholesterol Education Program (NCEP) Expert Panel on Cholesterol Levels in Children. Non-HDL-C values from the Bogalusa Heart Study are equivalent to the NCEP Pediatric Panel cut points for LDL-C. Values for plasma ApoB and ApoA-1 are from the National Health and Nutrition Examination Survey III (NHANES III). The threshold points for high and borderline-high values represent approximately the 95th and 75th percentiles, respectively. Low threshold points for HDL-C and ApoA-1 represent approximately the 10th percentile.

TC: total cholesterol; LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol; ApoB: apolipoprotein B; TG: triglycerides; ApoA-1: apolipoprotein A-1.

Reproduced from: Daniels SR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Full report, 2011. National Heart Lung and Blood Institute.

Available at: http://www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf.

Graphic 68183 Version 7.0

Characteristic findings of different genetic causes of hypercholesterolemia

Disorder	Estimated prevalence	Characteristic lipid levels*	Clinical features	
			In adults	In children
Heterozygous familial hypercholesterolemia	1/200 to 1/300	- Elevated TC and LDL-C - LDL-C 190 to 450 mg/dL[¶] - TG usually normal	- Tendon or tuberosus xanthomata - Xanthelasmas - Corneal arcus - Premature CHD	Subclinical atherosclerosis
Homozygous familial hypercholesterolemia	1/1,000,000	- Markedly elevated TC and LDL-C - Untreated LDL-C >500 mg/dL - TG usually normal	- Extensive xanthomata - Severe and progressive atherosclerotic cardiovascular disease	- Cutaneous tendon xanthomata often in the first year - Premature CHD occurs in childhood untreated
FCHL	1 to 2%	- Phenotypically heterogeneous - Elevated TC and/or TG - LDL-C/APOB ratio <1.2	- Xanthelasma - Corneal arcus - Premature CHD	Clinical manifestations are typically not seen during childhood
Familial hyperapobetalipoproteinemia	<1% Likely a variant of FCHL	- APOB >135 mg/dL - LDL-C <160 mg/dL - LDL-C/APOB ratio <1.2	- Xanthelasma - Premature CHD - Obesity	Clinical manifestations other than obesity are typically seen during childhood

Polygenic hypercholesterolemia	>25%	▪ Elevated TC and LDL-C ▪ TG usually normal ▪ LDL-C 130-250 mg/dL	▪ Premature CHD ▪ Tendon xanthomata are not seen	▪ Clinical manifest are typical not seen during childhood
Familial dysbetalipoproteinemia	1/5000	▪ Elevated TC and TG ▪ VLDL/TG ratio >0.3	▪ Tuberoeruptive xanthomata ▪ Xanthomata of the palmar creases ▪ Premature CHD	▪ Clinical manifest are typical not seen during childhood

Summary of the clinical features of different lipid disorders.

TC: total cholesterol; LDL-C: low-density lipoprotein cholesterol; TG: triglycerides; CHD: coronary heart disease; LDLR: LDL receptor; APOB: apolipoprotein B; PCSK9: proprotein convertase subtilisin kexin 9; FCHL: familial combined hyperlipidemia; VLDL: very low-density lipoprotein; APOE: apolipoprotein E.

* Lipid levels presented in this table represent the typical ranges seen in each disorder, primarily in affected adults. In most of these disorders, there is considerable variability in these values and lipid levels alone generally cannot be used to confirm or exclude the diagnosis.

¶ In pediatric patients, an LDL-C cutoff value of ≥ 190 mg/dL is generally used to establish a diagnosis of heterozygous FH. However, in patients with a concerning family history (ie, first-degree relatives with hypercholesterolemia and/or early cardiovascular disease), it may be appropriate to use a lower cutoff value (eg, LDL of ≥ 160 mg/dL).

Graphic 66747 Version 5.0

Risk stratification for children and adolescents at risk for early cardiovascular disease

High-risk conditions and risk factors	Moderate-risk conditions and risk factors	At-risk conditions and risk factors
▪ Homozygous FH ▪ Diabetes mellitus (type 1 or 2) ▪ End-stage kidney disease ▪ Kawasaki disease with persistent coronary aneurysms ▪ Solid-organ transplant vasculopathy ▪ Childhood cancer survivor following stem cell transplantation ▪ Multiple comorbidities – Any moderate-risk condition plus ≥ 2 additional moderate- or at-risk factors	▪ Severe obesity (BMI $\geq 99^{\text{th}}$ percentile or $\geq 35 \text{ kg/m}^2$) ▪ Confirmed hypertension (BP $>95^{\text{th}}$ percentile or $\geq 130/80 \text{ mmHg}$ on 3 separate occasions) ▪ Heterozygous FH ▪ Predialysis chronic kidney disease ▪ Aortic stenosis or coarctation ▪ Childhood cancer survivor with exposure to chest irradiation ▪ Multiple risk factors – ≥ 3 at-risk conditions or risk factors	▪ Obesity that is not severe (BMI $\geq 95^{\text{th}}$ to $<99^{\text{th}}$ percentile) ▪ Insulin resistance with comorbidities (eg, NAFLD, PCOS) ▪ Family history of premature CVD* ▪ Parent with known dyslipidemia (eg, FH) or TC $>240 \text{ mg/dL}$ (6.2 mmol/L) ▪ Current smoker or significant exposure to second-hand smoke ▪ White-coat hypertension (elevated BP measurements in the office with normal values outside the office setting) ▪ Chronic inflammatory disease (eg, SLE, systemic JIA) ▪ HIV infection ▪ Kawasaki disease with regressed coronary aneurysms ▪ Cardiomyopathy (eg, HCM) ▪ Surgically repaired congenital heart disease involving coronary artery translocation (eg, TGA repair)

		▪ Childhood cancer survivor with cardiotoxic chemotherapy only ▪ Adolescent depressive and bipolar disorders
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FH: familial hypercholesterolemia; BMI: body mass index; BP: blood pressure; NAFLD: nonalcoholic fatty liver disease; PCOS: polycystic ovary syndrome; CVD: cardiovascular disease; TC: total cholesterol; SLE: systemic lupus erythematosus; JIA: juvenile idiopathic arthritis; HCM: hypertrophic cardiomyopathy; TGA: transposition of the great arteries.

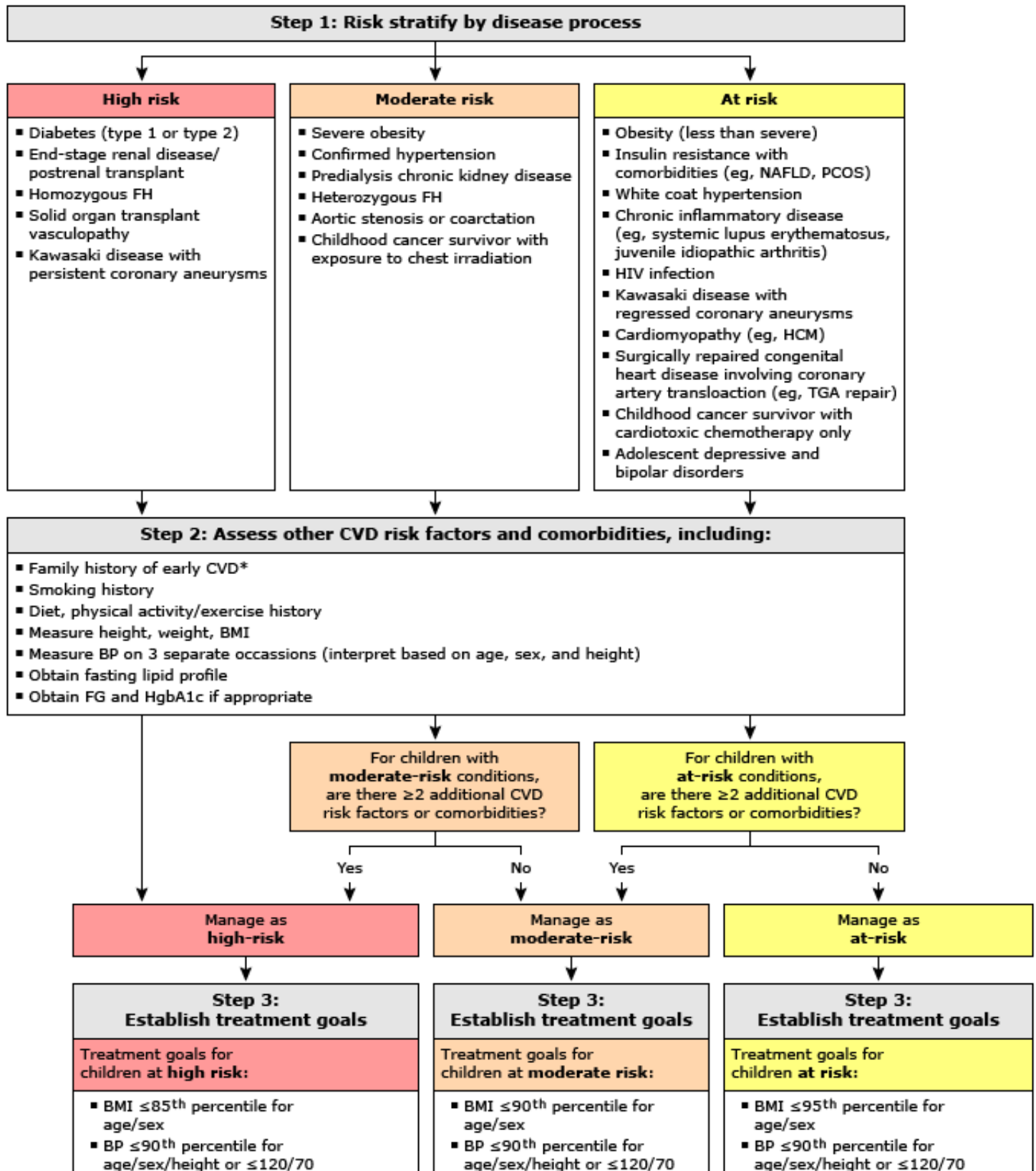
* Family history of premature CVD is generally defined as heart attack, treated angina, interventions for coronary artery disease, sudden cardiac death, or ischemic stroke in a first-degree relative (parent or sibling) before age 55 (males) or 65 (females).

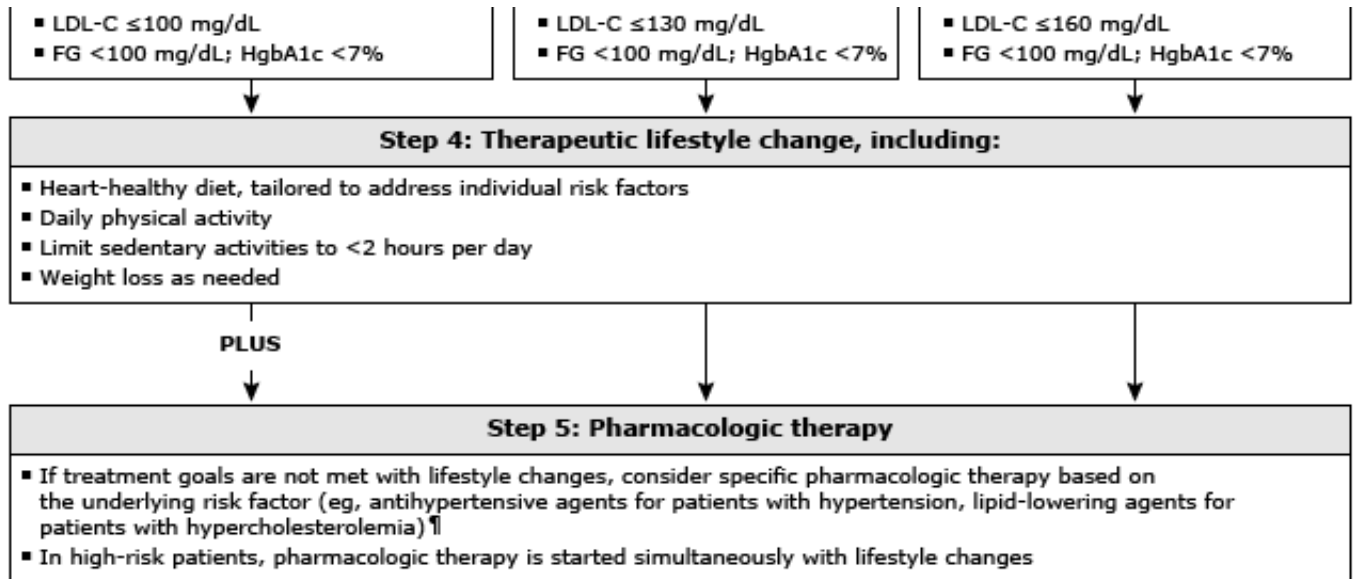
Adapted from:

1. Daniels SR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Full report, 2011. National Heart Lung and Blood Institute. Available at: http://www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf (Accessed on July 11, 2019).
2. de Ferranti SD, Steinberger J, Ameduri R, et al. Cardiovascular risk reduction in high-risk pediatric patients: A scientific statement from the American Heart Association. *Circulation* 2019; 139:e603.

Graphic 105366 Version 5.0

Risk stratification and management for children with conditions predisposing to accelerated atherosclerosis and early cardiovascular disease





FH: familial hypercholesterolemia; NAFLD: nonalcoholic fatty liver disease; PCOS: polycystic ovary syndrome; HCM: hypertrophic cardiomyopathy; TGA: transposition of the great arteries; CVD: cardiovascular disease; BMI: body mass index; BP: blood pressure; FG: fasting glucose; HgbA1c: hemoglobin A1c; LDL-C: low-density lipoprotein cholesterol.

* Family history of premature CVD is generally defined as heart attack, treated angina, interventions for coronary artery disease, sudden cardiac death, or stroke in a first-degree relative (parent or sibling) before age 55 (males) or 65 (females).

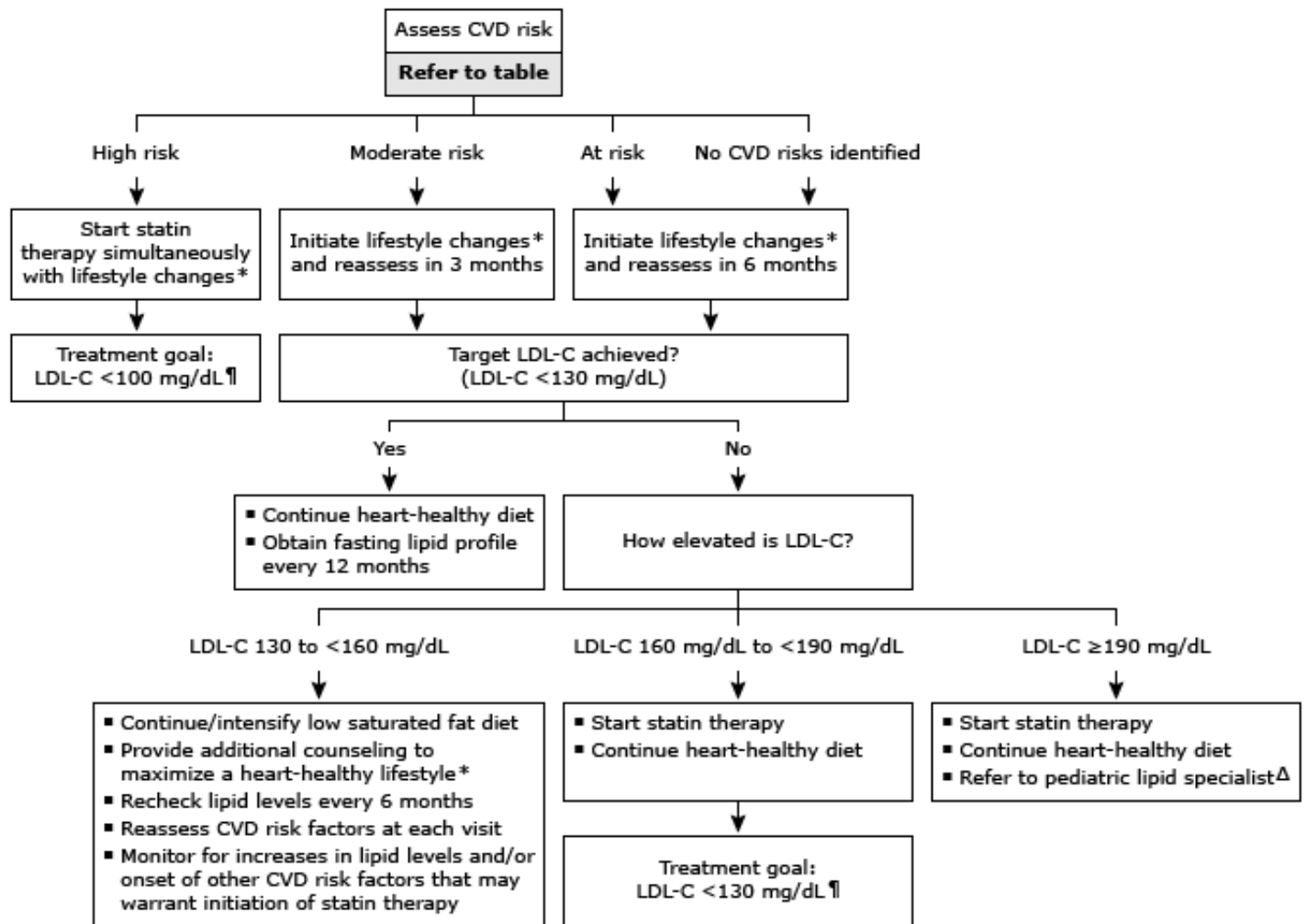
¶ For details of management of hypertension, dyslipidemia, and other specific conditions listed in this algorithm, refer to separate UpToDate content on these conditions in children.

Modified from:

1. Daniels SR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Full report, 2011. National Heart Lung and Blood Institute. Available at: www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf.
2. Goldstein BI, Carnethon MR, Matthews KA, et al. Major depressive disorder and bipolar disorder predispose youth to accelerated atherosclerosis and early cardiovascular disease: A scientific statement from the American Heart Association. *Circulation* 2015; 132:965.
3. De Ferranti SD, Steinberger J, Ameduri R, et al. Cardiovascular risk reduction in high-risk pediatric patients: A scientific statement from the American Heart Association. *Circulation* 2019; 139:e603.

Graphic 54963 Version 14.0

Overview of management of hypercholesterolemia in pediatric patients ≥ 10



This algorithm summarizes the management of children ≥ 10 years old with LDL-C levels in the range of treatment decisions should be based on averaging the LDL-C values from two separate fasting lipid profiles. Children <10 years old with clinically significant hypercholesterolemia and pediatric patients with LDL-C ≥ 190 mg/dL should be referred to a pediatric lipid specialist. This algorithm is intended for use in conjunction with additional UpToDate topics on the management of dyslipidemia in children for additional details of our approach to these treatments.

LDL-C units are mg/dL; divide by 38.67 to convert to mmol/L.

ASCVD: atherosclerotic cardiovascular disease; LDL-C: low density lipoprotein cholesterol; FH: familial hypercholesterolemia; NAFLD: nonalcoholic fatty liver disease; PCOS: polycystic ovary syndrome; TC: total cholesterol; SLE: systemic lupus erythematosus.

arthritis; HCM: hypertrophic cardiomyopathy; TGA: transposition of the great arteries; BP: blood pressure

* Lifestyle changes include:

- Diet modification: Encourage a diet high in fiber from fruits, vegetables, and whole grains; high in unsaturated fat; and devoid of trans fats; fat should comprise approximately 30% of total energy intake. Consultation with a registered dietitian may be warranted for implementation.
- Physical activity: Encourage at least 1 hour/day of moderate to vigorous activity; limit nonacademic screen time.
- Weight loss in obese children.
- Avoidance of nicotine exposure.

For additional details, refer to UpToDate content on pediatric dyslipidemia.

¶ For details on initiating, monitoring, and titrating statin therapy in children, refer to separate UpToDate content on pediatric dyslipidemia.

Δ Patients who continue to have LDL-C levels ≥ 190 mg/dL (4.9 mmol/L) despite initial lifestyle intervention should be referred to a pediatric lipid specialist. These patients are likely to have heterozygous FH or a similar genetic risk of premature ASCVD. Genetic testing may be warranted, and specialist care will often be beneficial. A pediatric cardiologist (or an endocrinologist, gastroenterologist, or general pediatrician) who has completed additional fellowship or via professional societies. If a pediatric lipid specialist is not available locally, referral to an adult lipid specialist for adolescents.

◇ Family history of premature ASCVD is generally defined as heart attack, treated angina, interventions for heart failure, death, or ischemic stroke in a first-degree relative (parent or sibling) before age 55 (males) or 65 (females).

Graphic 106010 Version 3.0

Dietary guidance for children older than 1 year

General guidelines
A variety of nutrient-dense foods from the basic food groups (cow milk/milk products or fortified soy products, meat/protein, grains, fruits/vegetables) should be offered each day.
Foods and beverages should contain or be prepared with little added salt, sugar, or caloric sweeteners.
Fat
1 to 2 years: Fat and cholesterol intake are not restricted.
2 to 3 years: Fat should comprise 30 to 40% of total energy intake; saturated fats should be limited to <10% of total energy intake; intake of trans fats should be as low as possible.
4 to 18 years: Fat should comprise 25 to 35% of total energy intake; fat intake should not be restricted to <20% of total energy intake; saturated fats should be limited to <10% of total energy intake; intake of trans fats should be as low as possible.
Meat/protein
Select and prepare meat, poultry, fish, and dried beans with as little fat as possible.
Fruits, vegetables, and fruit juice
A colorful variety of fruits and vegetables should be offered each day.
Whole fruit is preferred to fruit juice, but one-half of the recommended daily servings can be provided in the form of 100% fruit juice.
Consumption of 100% fruit juice should be limited to 4 ounces (120 mL) in children aged 1 through 3 years, 4 to 6 ounces (120 to 180 mL) in children aged 4 through 6 years, and 8 ounces (240 mL) in children older than 7 years.
Grains
At least one-half of total grains consumed should be whole grains. When reading the label, "whole grain" should be the first ingredient.
Cow milk or fortified soy milk
1 to 2 years: At least 2 cups (approximately 480 mL) of whole cow milk per day (or equivalent products).
2 to 8 years: At least 2 to 3 cups (approximately 480 to 720 mL) of fat-free or low-fat milk per

day (or equivalent products).
≥9 years: At least 3 cups (approximately 720 mL) of fat-free or low-fat milk per day (or equivalent products).
Beverages
Plain, unflavored water is the preferred beverage for children, particularly when fluids are consumed outside of meals and snacks.

Data from:

1. American Academy of Pediatrics Committee on Nutrition. Feeding the child. In: *Pediatric Nutrition*, 8th ed, Kleinman RE, Greer FR (Eds), American Academy of Pediatrics, Itasca, IL 2019. p.189.
2. USDA Scientific Report of the 2020 Dietary Guidelines Advisory Committee: Advisory Report to the Secretary of Agriculture and the Secretary of Health and Human Services. Available at: www.dietaryguidelines.gov/2020-advisory-committee-report (Accessed on August 03, 2020).
3. Gidding SS, Dennison BA, Birch LL, et al. Dietary recommendations for children and adolescents: A guide for practitioners: Consensus statement from the American Heart Association. *Circulation* 2005; 112:2061.
4. National Academies of Sciences, Engineering, and Medicine. Dietary reference intakes for calcium and vitamin D. Available at: <http://www.nationalacademies.org/hmd/Reports/2010/Dietary-Reference-Intakes-for-Calcium-and-Vitamin-D.aspx> (Accessed on December 12, 2019).
5. Heyman MB, Abrams SA, AAP Section on Gastroenterology, Hepatology, and Nutrition, Committee on Nutrition. Fruit juice in infants, children, and adolescents: Current recommendations. *Pediatrics* 2017; 139:e20170967.
6. Lott M, Callahan E, Welker Duffy E, et al. Healthy beverage consumption in early childhood: Recommendations from key national health and nutrition organizations. Technical Scientific Report. Healthy Eating Research, Durham, NC 2019. Available at: <https://healthydrinkshealthykids.org/professionals/> (Accessed on October 09, 2019).

Graphic 79084 Version 18.0

Sources of dietary fatty acids

Type of fat	Food sources
Monounsaturated fatty acids	Vegetable oils (canola, olive, sunflower, safflower), peanut, tree nuts, seeds, avocado
Polyunsaturated fatty acids; n-3 or omega-3	ALA – Vegetable oils (canola, soybean, walnut), flax/linseed/hemp chia seed, wheat germ; also vegetables of the cabbage family and some fortified eggs and fortified foods EPA and DHA – Seafood (especially fatty fish); some infant formulas, fortified eggs, and fortified foods
Polyunsaturated fatty acids; n-6 or omega-6	LA – Vegetable oils (soybean, corn, cottonseed), peanut, tree nuts, seeds, other vegetable sources, poultry
Saturated fatty acids	Full-fat or fat-reduced dairy products, meat, poultry, vegetable oils (coconut, palm kernel, palm)
Trans fatty acids	Partially hydrogenated vegetable oils (stick and full-fat margarine, commercial baked goods, deep-fried foods)

ALA: alpha-linolenic acid; EPA: eicosapentaenoic acid; DHA: docosahexaenoic acid; LA: linoleic acid.

Graphic 80214 Version 9.0

Activity recommendations for cardiovascular health from the Expert Panel on Integrated Guidelines for Cardiovascular Health and Risk Reduction in Children and Adolescents

Age	Activity recommendations
0 to 12 months	Parents should create an environment promoting and modeling physical activity and limiting sedentary time
	<i>Supportive actions:</i> Discourage TV viewing altogether
1 to 4 years	Unlimited active playtime in safe, supportive environment
	Limit sedentary time, especially TV/video
	<i>Supportive actions:</i> - For children <2 years, discourage TV viewing altogether - Limit total media time to no more than 1 to 2 hours of quality programming per day - No TV in child's bedroom - Encourage family activity at least once per week - Counsel routine activity for parents as role models for children
5 to 10 years	Moderate to vigorous physical activity* every day
	Limit daily leisure screen time (TV/video/computer)
	<i>Supportive actions:</i> - Prescribe moderate to vigorous activity* 1 hour/day with vigorous intensity physical activity[¶] on 3 days/week - Limit total media time to no more than 1 to 2 hours of quality programming per day - No TV in child's bedroom - Take activity and screen time history from child once a year - Match physical activity recommendations with energy intake - Recommend appropriate safety equipment relative to each sport - Support recommendations for daily physical education in schools
11 to 17 years	Moderate to vigorous physical activity* every day
	Limit leisure time TV/video/computer use
	<i>Supportive actions:</i>

	- Encourage adolescents to aim for 1 hour/day of moderate to vigorous daily activity*, with vigorous intense physical activity¶ on 3 days/week - Encourage no TV in bedroom - Limit total media time to no more than 1 to 2 hours of quality programming per day - Match activity recommendations with energy intake - Take activity and screen time history from adolescent at health supervision visits - Encourage involvement in year-round, physical activities - Support continued family activity once a week and/or family support of adolescent's physical activity program - Endorse appropriate safety equipment relative to each sport
18 to 21 years	Moderate to vigorous physical activity* every day
	Limit leisure time TV/video/computer
	<i>Supportive actions:</i> - Support goal of 1 hour/day of moderate to vigorous daily activity with vigorous intense physical activity on 3 days/week - Recommend that combined leisure screen time not exceed 2 hours/day - Activity and screen time history at health supervision visits - Encourage involvement in year-round, lifelong physical activities

TV: television.

* Examples of moderate to vigorous physical activities are jogging or playing baseball.

¶ Examples of vigorous physical activities are running, playing singles tennis, or playing soccer.

Modified from: Expert Panel on Integrated Guidelines for Cardiovascular Health and Risk Reduction in Children and Adolescents: Summary Report. Available at: <https://www.nhlbi.nih.gov/node/80139> (Accessed on January 3, 2012).

Graphic 75748 Version 6.0

Lipid-lowering medications in children

Class	Drug	Initial daily dose	Upper daily dose studied in clinical trials (children aged ≥ 10 years) ^[1]	Maximum daily dose (adult)	Available strengths (United States)
Statins					
	Atorvastatin	5 to 10 mg	20 mg	80 mg	10, 20, 40, 80 mg
	Fluvastatin	20 mg	80 mg	80 mg	20, 40 mg, 80 mg (extended release)
	Lovastatin	10 mg	40 mg	80 mg	10, 20, 40 mg 20, 40, 60 mg (extended release)
	Pravastatin	10 mg	40 mg (age 14 to 18 years) 20 mg (age 8 to 13 years)	80 mg	10, 20, 40, 80 mg
	Rosuvastatin	5 mg	20 mg	40 mg	5, 10, 20, 40 mg
	Simvastatin	10 mg (age ≥ 10 years) 5 mg (age 10 years)	40 mg	40 mg	5, 10, 20, 40, 80 [¶] mg
Cholesterol absorption inhibitors					
	Ezetimibe	10 mg	10 mg	10 mg	10 mg

Fibric acid derivatives					
	Gemfibrozil	1200 mg [*]	Data lacking	1200 mg	600 mg
	Fenofibrate	40 mg	Data lacking	130 to 200 mg (maximum dose depends on preparation; see Lexicomp drug reference)	40, 43, 48, 50, 54, 67, 100, 120, 134, 145, 160, 200 mg (available strengths vary by preparation; see Lexicomp drug reference)
Bile acid sequestrants					
	Cholestyramine	2 to 4 g or 240 mg/kg per day	8 g 4 g (age <10 years)	16 g	4 g per packet or scoop
	Colestipol	2.5 to 5 g	12 g	20 g	5 g per packet or scoop 1 g tablet
	Colesevelam	1.25 g	3.75 g	3.75 g	3.75 g per packet 625 mg tablet
Omega-3 fish oils					
	Omega-3 acid ethyl esters	1 g	Data lacking	4 g	1 g (many strengths available)

Always review drug interactions upon initiation and alteration of treatment. Specific interactions may be determined using the drug interactions tool (Lexi-Interact Online) included with UpToDate. This tool can be accessed from the UpToDate online search page or through the individual drug information topics in the section on Drug interactions.

Children and adolescents are generally **not** managed with more than one class of lipid-lowering agents due to increased risk of side effects. Combinations of statins and fibrates should be administered under the supervision of a pediatric lipid specialist.

LDL-C: low-density lipoprotein cholesterol.

* An initial 600 mg daily dose of gemfibrozil may be considered. Data in children are lacking.

¶ The US Food and Drug Administration has placed a black box label on simvastatin at a dose of 80 mg per day because it is associated with muscle toxicity in adults. If a daily dose of 40 mg of simvastatin is insufficient to achieve LDL-C control, a change to a more potent statin (eg, atorvastatin or rosuvastatin) is suggested. Doses of simvastatin that exceed 40 mg per day should be given under the supervision of a pediatric lipid specialist, and are generally not required, as more potent statins are available.

Reference:

1. Data from: Daniels DR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: summary report-2011. National Heart Lung and Blood Institute. Available at: <https://www.nhlbi.nih.gov/node/80139> (Accessed April 4, 2012).

Graphic 69650 Version 10.0

Cytochrome P450 3A (including 3A4) inhibitors and inducers

Strong inhibitors	Moderate inhibitors	Strong inducers	Moderate inducers
- Adagrasib - Atazanavir - Ceritinib - Clarithromycin - Cobicistat and cobicistat-containing coformulations - Darunavir - Idelalisib - Indinavir - Itraconazole - Ketoconazole - Levoketoconazole - Lonafarnib - Lopinavir - Mifepristone* - Nefazodone - Nelfinavir - Nirmatrelvir-ritonavir - Ombitasvir-paritaprevir-ritonavir - Ombitasvir-paritaprevir-ritonavir plus dasabuvir - Posaconazole - Ritonavir and ritonavir-containing	- Amiodarone[¶] - Aprepitant - Berotrastat - Cimetidine[¶] - Conivaptan - Crizotinib - Cyclosporine[¶] - Diltiazem - Duvelisib - Dronedarone - Erythromycin - Fedratinib - Fluconazole - Fosamprenavir - Fosaprepitant[¶] - Fosnetupitant-palonosetron - Grapefruit juice - Imatinib - Isavuconazole (isavuconazonium sulfate) - Lefamulin - Letermovir - Netupitant - Nilotinib - Ribociclib - Schisandra - Verapamil	- Apalutamide - Carbamazepine - Enzalutamide - Fosphenytoin - Lumacaftor - Lumacaftor-ivacaftor - Mitotane - Phenobarbital - Phenytoin - Primidone - Rifampin (rifampicin)	- Bexarotene - Bosentan - Cenobamate - Dabrafenib - Dexamethasone^Δ - Dipyrrone - Efavirenz - Elagolix, estradiol, and norethindrone therapy pack[◇] - Eslicarbazepine - Etravirine - Lorlatinib - Mitapivat - Modafinil - Nafcillin - Pexidartinib - Rifabutin - Rifapentine - Sotorasib - St. John's wort

coformulations			
▪ Saquinavir			
▪ Tucatinib			
▪ Voriconazole			

- For drug interaction purposes, the inhibitors and inducers of CYP3A metabolism listed above can alter serum concentrations of drugs that are dependent upon the CYP3A subfamily of liver enzymes, including CYP3A4, for elimination or activation.
- These classifications are based upon US Food and Drug Administration (FDA) guidance.^[1,2] Other sources may use a different classification system resulting in some agents being classified differently.
- Data are for systemic drug forms. Degree of inhibition or induction may be altered by dose, method, and timing of administration.
- Weak inhibitors and inducers are not listed in this table with exception of a few examples. Clinically significant interactions can occasionally occur due to weak inhibitors and inducers (eg, target drug is highly dependent on CYP3A4 metabolism and has a narrow therapeutic index). Accordingly, specific interactions should be checked using a drug interaction program such as the [Lexicomp drug interactions](#) program included within UpToDate.
- Refer to UpToDate topics on specific agents and indications for further details.

* Mifepristone is a significant inhibitor of CYP3A4 when used chronically (eg, for hyperglycemia in patients with Cushing syndrome); not in single-dose use.

¶ Classified as a weak inhibitor of CYP3A4 according to FDA system.^[1]

Δ Classified as a weak inducer of CYP3A4 according to FDA system.^[1]

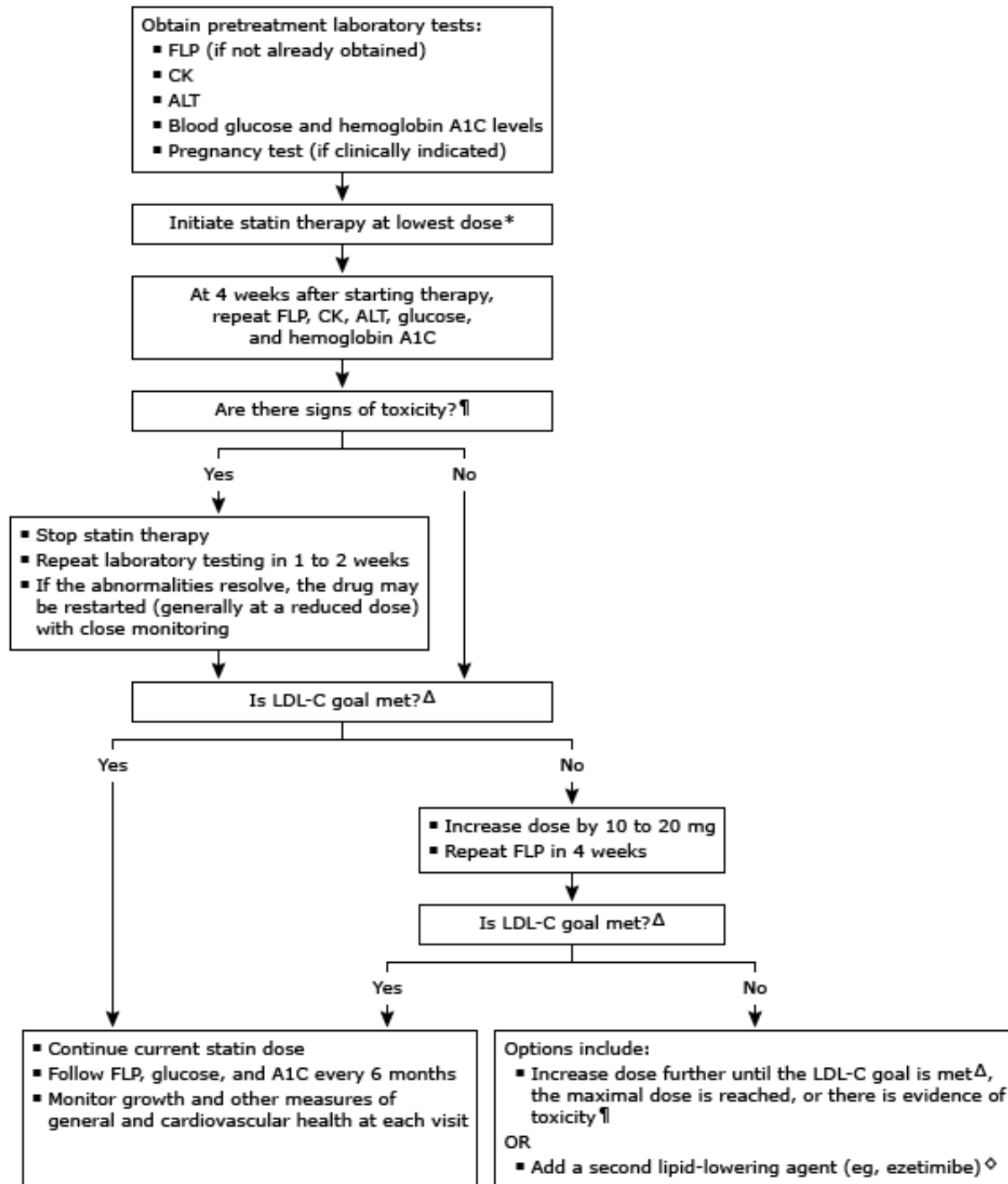
◇ The fixed-dose combination therapy pack taken in the approved regimen has moderate CYP3A4 induction effects. When elagolix is used as a single agent, it is a weak CYP3A4 inducer. Norethindrone and estradiol are not CYP3A4 inducers.

Data from: Lexicomp Online (Lexi-Interact). Copyright © 1978-2023 Lexicomp, Inc. All Rights Reserved.

References:

1. Clinical Drug Interaction Studies — Cytochrome P450 Enzyme- and Transporter-Mediated Drug Interactions Guidance for Industry (January 2020) available at: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-drug-interaction-studies-cytochrome-p450-enzyme-and-transporter-mediated-drug-interactions>.
2. US Food & Drug Administration. Drug Development and Drug Interactions: Table of Substrates, Inhibitors and Inducers. Available at: [FDA.gov website](#).

Overview of statin therapy for hypercholesterolemia in pediatric patients ≥ 10 years old



This figure summarizes the management of statin therapy in children and adolescents 10 years and older. Children <10 years who require statin therapy should be managed by a pediatric lipid specialist. Prior to initiating statin therapy, potential drug

interactions should be reviewed if the child is taking other medications (specific drug interactions may be determined by using Lexi-Interact). Adolescent female patients should be advised about concerns with pregnancy and the need for appropriate contraception. This algorithm is intended for use in conjunction with additional UpToDate content. Refer to UpToDate topics on pediatric dyslipidemia for additional details.

FLP: fasting lipid profile; CK: creatinine kinase; ALT: alanine aminotransferase; LDL-C: low density lipoprotein cholesterol; CVD: cardiovascular disease.

* Refer to drug information topics for information on dosing of individual statin medications in children. The initial choice of agent is generally based on potential drug interactions, experience with the drug in children, price, and patient preference. Statin medications are usually given at bedtime, because most LDL-C synthesis occurs during nighttime hours.

¶ Adverse side effects from statin therapy are uncommon. Toxicity can include elevated liver enzymes (≥ 3 times the upper limit of normal) and myopathy (ie, elevated CK [≥ 10 times the upper limit of normal] or suggestive symptoms such as weakness, asthenia, and/or muscle cramps). These are more likely at higher doses and in patients taking other medications. Routine monitoring of CK and ALT is suggested at 4 weeks after starting therapy. Repeated testing of liver function and CK in asymptomatic patients is generally not necessary. However, testing should be performed if the patient has concerning symptoms (eg, muscle aches or weakness) or other comorbidities (eg, liver disease).

Δ Targeted values of LDL-C depend on the presence of associated CVD risk factors. For children at high risk, the goal is LDL-C <100 mg/dL (2.6 mmol/L); for children in the moderate- and at risk categories, the goal is LDL-C <130 mg/dL (3.36 mmol/L).

◇ Care must be taken if a fibrate is added to a statin because this may increase the risk of side effects, particularly muscle toxicity. Referral to a pediatric lipid specialist is advised.

Graphic 106011 Version 3.0

